EXPERIMENT 7

SPECTRAL ESTIMATION

7.1 Power Spectral Density Estimation and Spectral Analysis of Noisy Sinusoidal Signals Using Periodogram Method in MATLAB

Program (Matlab code):

```
clc; clear; close all
fs=1000; t=(0:999)/fs;
s=sin(2*pi*100*t)+0.5*sin(2*pi*200*t);
n=0.3*randn(size(t));
x=s+n;
figure, plot(x), grid on
title('Signal with Noise')
figure, periodogram(x,[],[],fs)
title('PSD using Periodogram')
figure
X=abs(fft(x));
f=(0:length(X)-1)*fs/length(X);
plot(f(1:500),X(1:500)), grid on
title('Magnitude Spectrum')
figure
bar([mean(s.^2) var(n)])
set(gca,'XTickLabel',{'Signal','Noise'})
title('Power Comparison')
grid on
```

OUTPUT

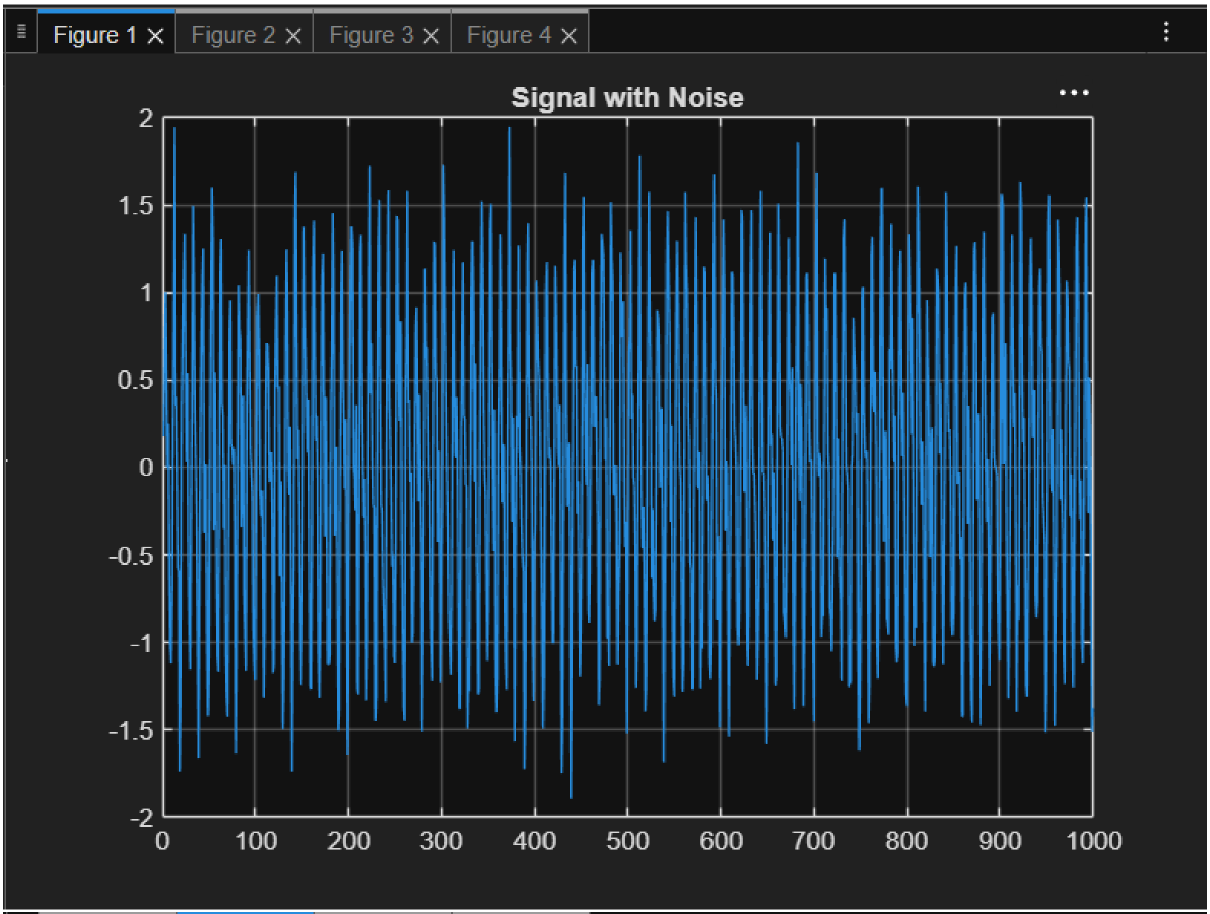


Figure 1
Figure 2
Figure 3
Figure 4
Signal with Noise
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0
100
200
300
400
500
600
700
800
900
1000

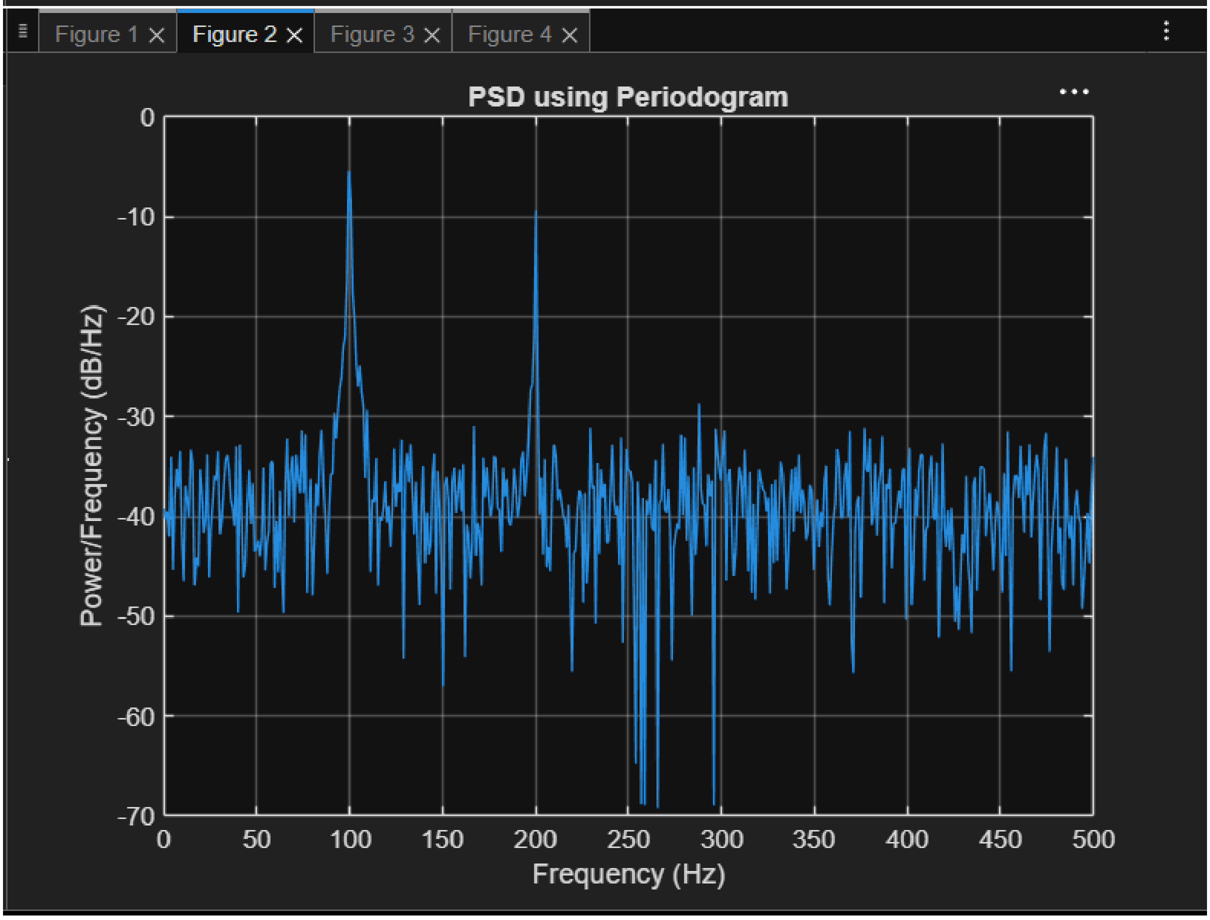


Figure 1
Figure 2
Figure 3
Figure 4
PSD using Periodogram
Power/Frequency (dB/Hz)
0
-10
-20
-30
-40
-50
-60
-70
0
50
100
150
200
250
300
350
400
450
500
Frequency (Hz)

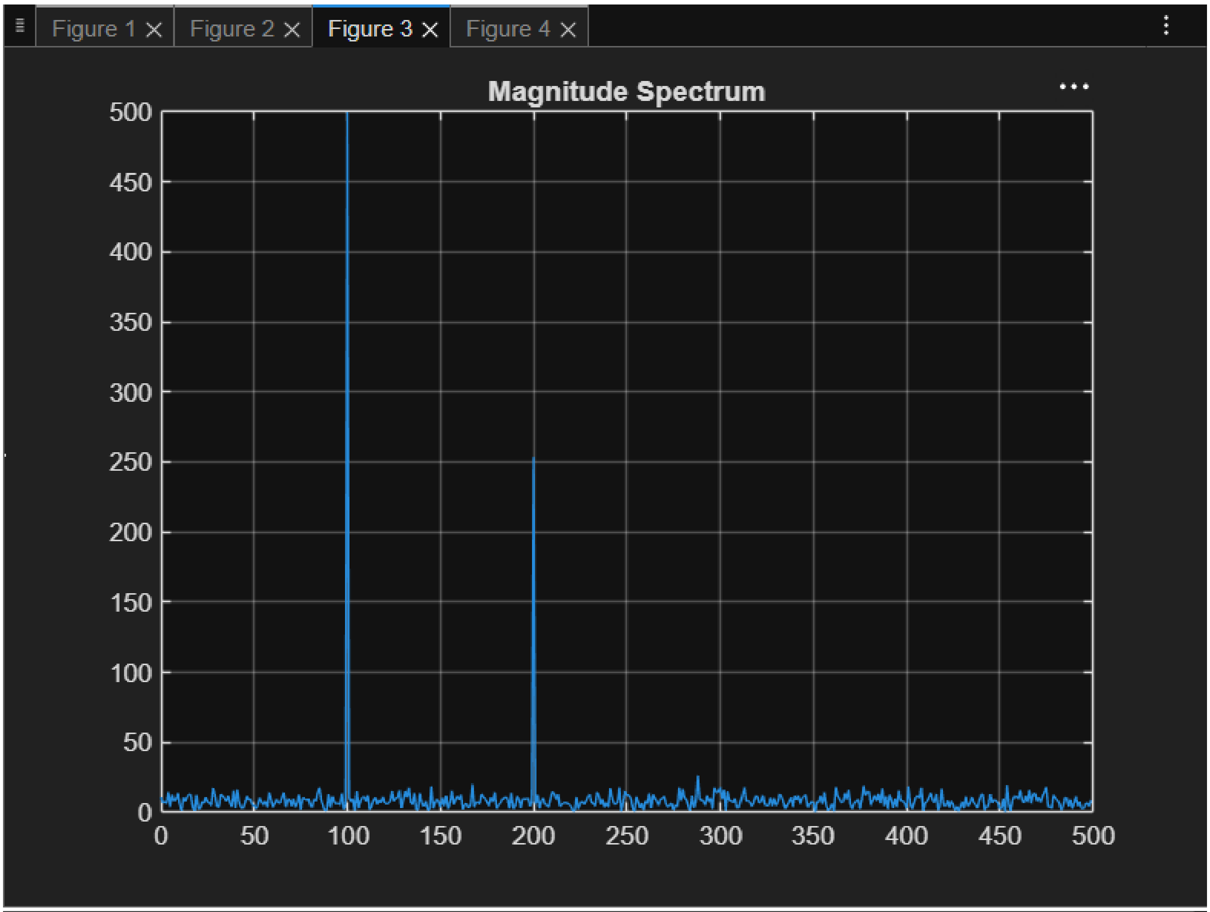


Figure 1
Figure 2
Figure 3
Figure 4
Magnitude Spectrum
500
450
400
350
300
250
200
150
100
50
0
0
50
100
150
200
250
300
350
400
450
500

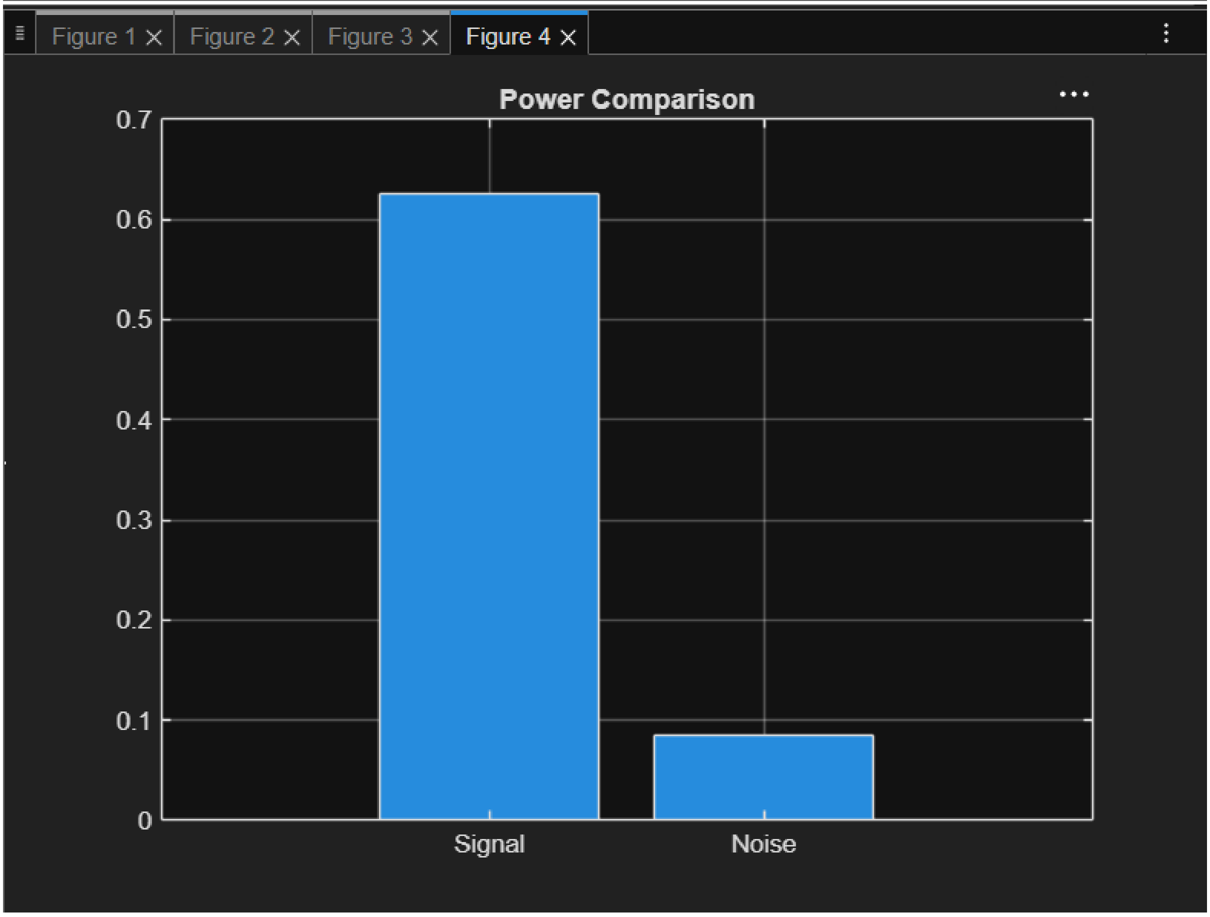


Figure 1
Figure 2
Figure 3
Figure 4
Power Comparison
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
Signal
Noise

7.2 Analysis of Windowing Effects on Spectral Estimation Using Rectangular, Hamming, and Hanning Windows in MATLAB

PROGRAM

```
clc; clear; close all;

% Signal
fs = 1000; N = 256; t = (0:N-1)/fs;
x = sin(2*pi*100*t) + 0.8*sin(2*pi*110*t);
NFFT = 2048; f = (0:NFFT/2-1)*(fs/NFFT);

% Windows
w = {rectwin(N), hamming(N), hann(N)};
names = {'Rectangular','Hamming','Hanning'};

% Processing
for i = 1:3
xw = x(:).*w{i};
X = fft(xw,NFFT);
P{i} = abs(X(1:NFFT/2));
P{i} = P{i}/max(P{i});

end

%% Individual Spectrums
figure;
for i = 1:3
subplot(3,1,i);
plot(f,20*log10(P{i}),'LineWidth',1.5);
title(['Spectrum using ' names{i} ' Window']);
xlim([0 250]); grid on;
```

```
end

%% Comparison
figure;
plot(f,20*log10(P{1}),'k',f,20*log10(P{2}),'r',f,20*log10(P{3}),'b','LineWidth',1.5);
legend(names); title('Window Comparison'); xlim([50 170]); grid on;
%% Window Shapes
figure;
for i = 1:3
subplot(3,1,i); plot(w{i},'LineWidth',1.5);
title([names{i} ' Window']); grid on;
end
%% Peak Detection
for i = 1:3
[pks{i},loc{i}] = findpeaks(P{i},f,'MinPeakHeight',0.3);
fprintf('\n%s Window Frequencies:\n',names{i});
disp(loc{i});
end
%% Peak Marking (Hamming)
figure;
plot(f,20*log10(P{2}),'LineWidth',1.5); hold on;
plot(loc{2},20*log10(pks{2}),'ro','MarkerFaceColor','r');
title('Detected Frequencies (Hamming)'); xlim([50 170]); grid on;
%% Main Peaks
fprintf('\nMain Spectral Peaks:\n');
for i = 1:3
[~,idx] = max(P{i});
fprintf('%s Window Peak = %.2f Hz\n',names{i},f(idx));
end
```

OUTPUT

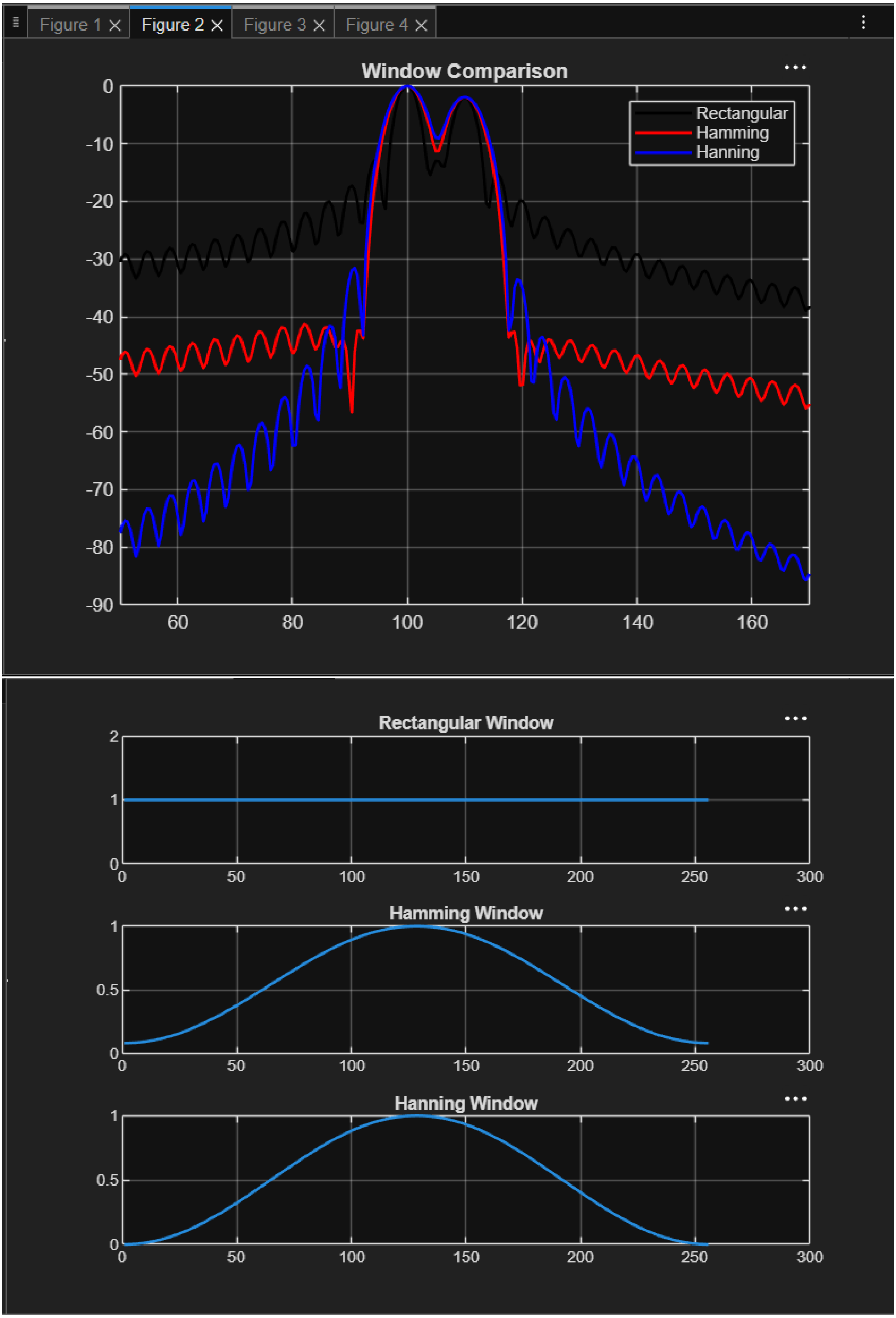
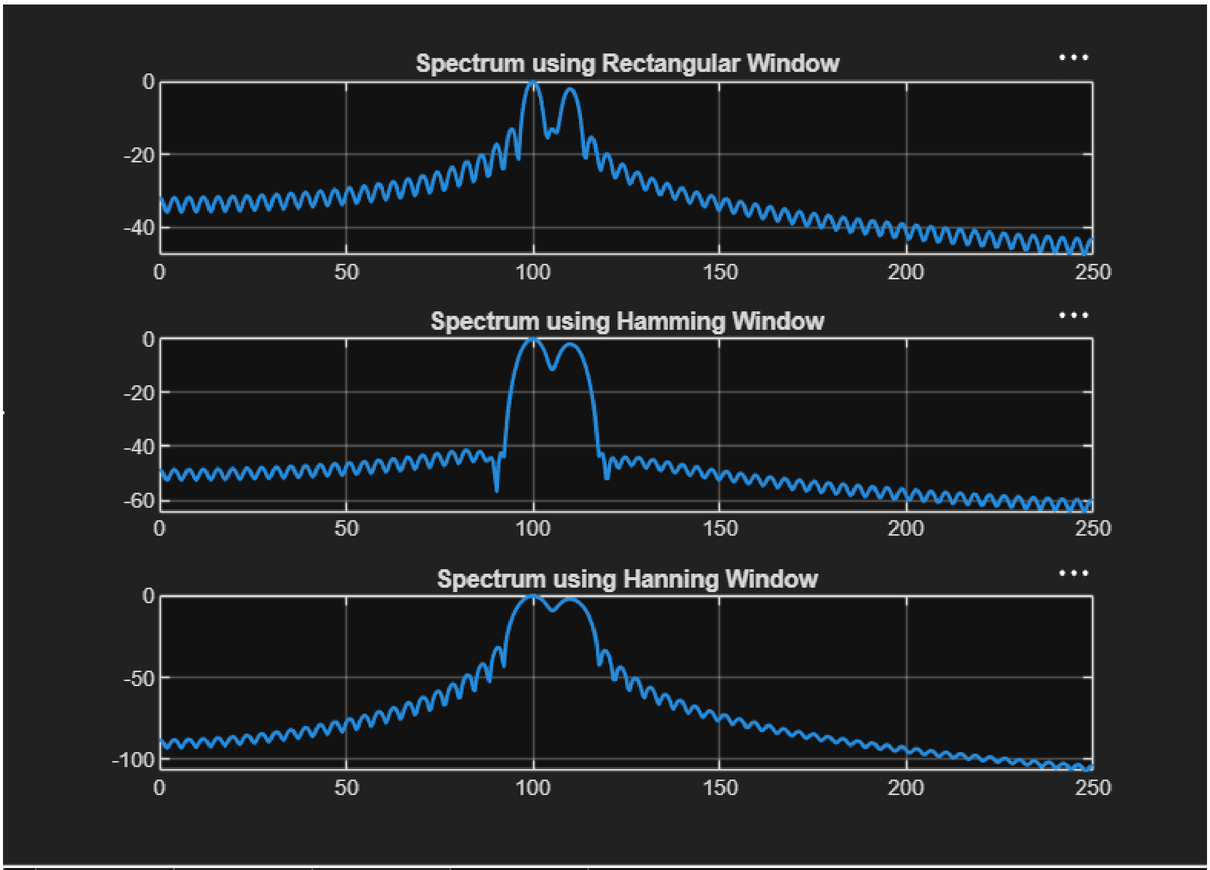
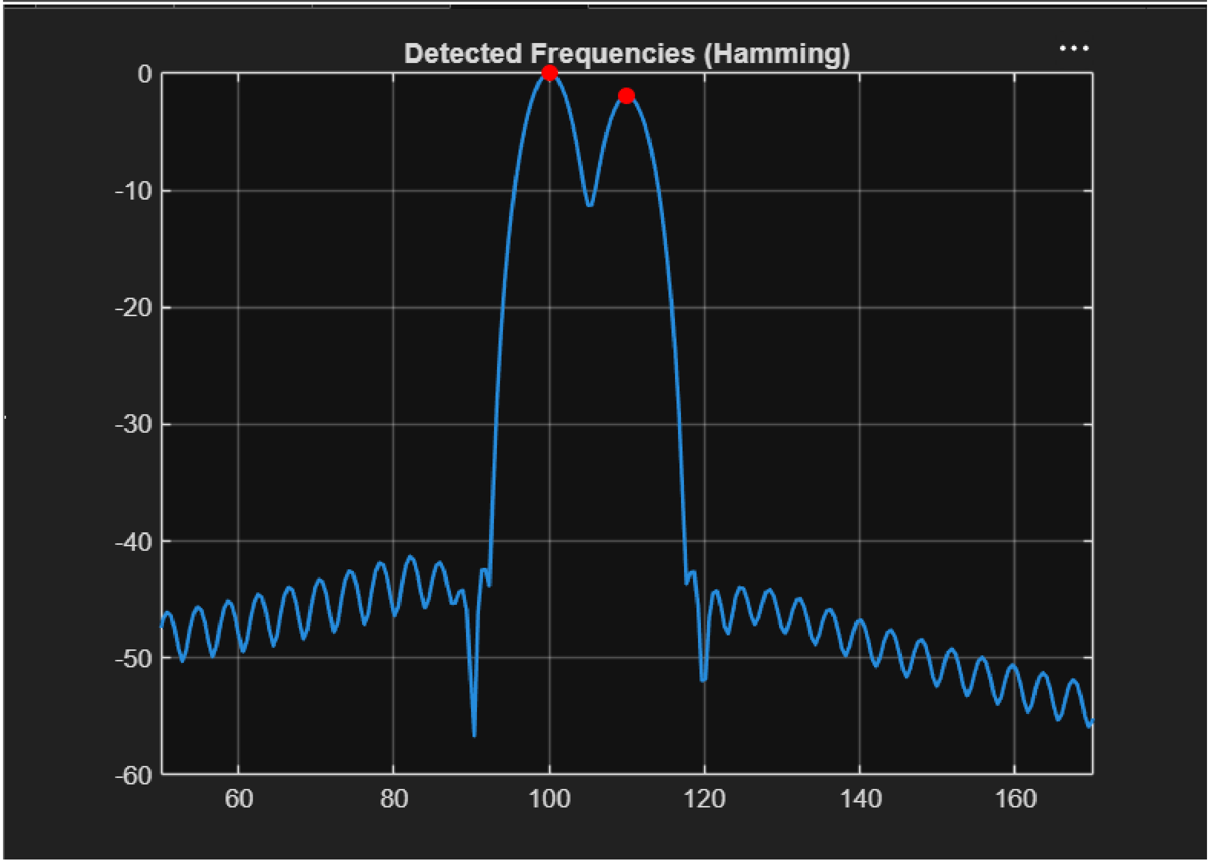


Figure 1
Figure 2
Figure 3
Figure 4
Window Comparison
Rectangular
Hamming
Hanning
0
-10
-20
-30
-40
-50
-60
-70
-80
-90
60
80
100
120
140
160
Rectangular Window
Hamming Window
Hanning Window
2
1
0
0.5
50
150
200
250
300


Spectrum using Rectangular Window
0
-20
-40
0
50
100
150
200
250
Spectrum using Hamming Window
0
-20
-40
-60
0
50
100
150
200
250
Spectrum using Hanning Window
0
-50
-100
0
50
100
150
200
250

Detected Frequencies (Hamming)
0
-10
-20
-30
-40
-50
-60
60
80
100
120
140
160